THE VISUAL ATLAS OF REALITY

Wave-Substrate Theory: A Geometric Foundation for Physics

PRIOR ART UPDATE

Trefoil Knot Models of the Proton: What Existed Before WST

Supplement to the Discovery Priority & Prior Art Index

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Purpose

This document updates the Prior Art Index (Item #11: Proton as (2,3) Torus Knot) with three additional precursors identified during adversarial review. The qualitative idea of the proton as a trefoil knot standing wave has been explored by multiple independent researchers.

What distinguishes Wave-Substrate Theory is the quantitative derivation program: 13+ quantities derived from the trefoil topology with zero free parameters, matching experiment from exact agreement to ~1%. No prior framework achieves this. The prior art strengthens the WST priority claim by demonstrating convergent identification of the trefoil topology from independent sources, while clearly delineating the boundary between qualitative proposal and quantitative theory.

> *The distinction between qualitative proposal and quantitative theory is the distinction between suggesting the proton might be a trefoil knot and proving that a trefoil knot produces the measured proton-electron mass ratio to 0.13σ.*

Previously Acknowledged Prior Art

The following precursors are already documented in the Priority Index and book manuscript (Chapter 23, Appendix D):

Author / Year	What Was Proposed	What Was NOT Proposed	Status in Priority Index
Lord Kelvin (1867)	Atoms as knotted vortices in the ether. Topological intuition correct.	No specific knot-to-particle mapping. Abandoned after ether disproof.	Acknowledged — qualitative pioneer
Tony Skyrme (1961)	Baryon number as topological winding number (Skyrmions). Quantitative for nuclear physics.	Does not identify specific knot types or derive fundamental constants.	Acknowledged — topological soliton model
Faddeev & Niemi (1997)	Stable knotted solitons in modified Yang-Mills theory.	Mathematical physics; not applied to specific particle properties.	Acknowledged — knotted solitons
Buniy & Kephart (2005)	Glueball masses from knot energies applied to QCD bound states.	Different target (glueballs, not protons); different method (knot energy, not standing-wave DOF counting).	Acknowledged — knot energy calculations

Newly Identified Prior Art

Three precursors identified during adversarial review (March 2026). Each should be added to Priority Index Item #11.

1. Jack S. Avrin (2012)

Knots on a Torus: A Model of the Elementary Particles — Symmetry 4(1), 39–80 (2012). Peer-reviewed, open access (MDPI).

What Avrin Proposed

A complete particle taxonomy built from the unknot and trefoil as localized distortions of a featureless continuum. Fermions and bosons, hadrons and leptons, spin, isospin, mass, charge, CPT invariance, and three generations all emerge from knot topology. Avrin explicitly uses unknot = electron, trefoil = proton, identifies three-fold structure with quark correspondence, derives charge from topology, and obtains three generations. The philosophical framework — 'particles as localized distortions of a continuum' — is essentially identical to WST's substrate ontology.

What Avrin Did NOT Have

No mass ratio derivation. No $6\pi^5$. No charge radius prediction. No CKM parameters. No neutron-proton mass difference formula. No self-energy correction program. No quantitative predictions matching experiment to sub-sigma precision. The work is qualitative topology, not quantitative geometry.

PRIOR ART: Avrin (2012): Significant structural overlap — same two knots, same philosophical framework. Zero quantitative predictions. Strengthens the case that the topology is physically motivated.

WST ORIGINAL: **Recommended action.** Add to Priority Index Item #11 prior art after Buniy & Kephart (2005). Cite in Chapter 23 alongside Kelvin and Skyrme.

2. John Duffield (2016–2020)

'The Proton,' physicsdetective.com (2016, updated 2020). Blog post, not peer-reviewed.

What Duffield Proposed

The proton as 'a spherically-symmetric three-loop trefoil knot of field-variation configured as a standing-wave particle.' Electron = photon in a trivial-knot configuration; proton = photon in a trefoil-knot configuration. Duffield independently identifies: (a) proton = trefoil standing wave, (b) electron = unknot, (c) mass ratio ~1837 as a consequence of the topology, (d) charge as topological, (e) quarks as loops within the knot, (f) confinement as topological impossibility of unknotting. The 'goes round six times' structure is noted. Standing-wave language is nearly identical to WST's.

What Duffield Did NOT Have

No derivation of the mass ratio — states 1837 as a fact to be explained, not as a consequence of $6\pi^5$. No DOF counting, no D_3 symmetry factor, no standing-wave integral, no charge radius, no CKM, no neutron mechanism. The 'goes round six times' observation is noted but not connected to the dihedral group. No mathematical framework whatsoever — purely qualitative and pictorial.

PRIOR ART: Duffield (2016): Striking conceptual overlap — nearly identical standing-wave language, same topology, same mass ratio observation. Zero mathematical derivation. Duffield had the picture. WST has the proof.

WST ORIGINAL: **Recommended action.** Add to Priority Index Item #11 as independent qualitative proposal. Note in Chapter 23 as convergent intuition from an independent researcher.

3. Greg Vit (2023)

'A Geometric Derivation of the Electron, Proton, and Neutron Magnetic Moments,' HAL open archive hal-04091300 (2023). Preprint, not peer-reviewed.

What Vit Derived

Magnetic moments for the electron (~Bohr magneton), proton (~99.66% of measured), and neutron from geometric models. Electron = two-loop solenoid (unknot topology). Proton and neutron = current-carrying wires folded into a trefoil knot. Vit explicitly states the moments 'may be better represented as the result of oscillating standing waves or rotating charges.' This is the closest prior art to WST's quantitative program — the same topology used to produce a quantitative prediction for a physical observable.

What Vit Did NOT Have

No mass ratio derivation. No charge radius. No CKM parameters. No neutron-proton mass difference from phase inversion. No self-energy program. One quantity (magnetic moment) at ~99.5%, not 13 quantities at sub-sigma. Classical current-loop model, not standing-wave DOF counting on the knot manifold. No substrate ontology.

PRIOR ART: Vit (2023): Closest prior art to WST's quantitative program. Uses trefoil geometry to predict one observable (~99.5%). WST extends to 13+ observables across independent domains using an entirely different method.

WST ORIGINAL: **Recommended action.** Add to Priority Index Item #11 prior art. Cite as independent quantitative work using trefoil geometry.

Comparison: Prior Art vs. WST

Capability	Kelvin 1867	Skyrme 1961	Avrin 2012	Duffield 2016	Vit 2023	WST 2026
Proton = trefoil knot	—	—	✓	✓	✓	✓
Electron = unknot	—	—	✓	✓	✓	✓
Standing-wave model	—	—	—	✓	Noted	✓
Quarks = lobes	—	—	✓	✓	✓	✓
Confinement = topology	✓	✓	✓	✓	—	✓
Three generations	—	—	✓	—	—	✓
Mass ratio m_p/m_e derived	—	—	—	—	—	✓ ($6\pi^5$, 0.13σ)
Charge radius r_p derived	—	—	—	—	—	✓ (< 1σ)
Neutron = phase inversion	—	—	—	—	—	✓
Δm(n−p) derived	—	—	—	—	—	✓ (0.2%)
CKM parameters derived	—	—	—	—	—	✓
Magnetic moments	—	—	—	—	✓ (99.5%)	✓ (g−2 program)
Self-energy program (c_2,c_3)	—	—	—	—	—	✓
# quantities derived	0	0	0	0	1	13+
Free parameters	N/A	N/A	N/A	N/A	N/A	0

Attribution Statement

The qualitative identification of the proton as a (2,3) torus knot (trefoil) standing wave is not unique to Wave-Substrate Theory. Multiple independent researchers have proposed this topology: Avrin (2012) in a peer-reviewed journal, Duffield (2016–2020) in public blog posts, and Vit (2023) in a preprint achieving ~99.5% accuracy on magnetic moments.

WST acknowledges all three as independent prior art and notes that their independent convergence on the same topology from different directions provides additional evidence that the trefoil identification is physically correct, not arbitrary.

What is original to WST is the comprehensive quantitative derivation program: a suite of 13+ derived quantities matching experiment with zero free parameters. No prior framework — qualitative or quantitative — achieves this scope or precision.

Recommended Changes to Priority Index

Item #11 (Proton as (2,3) Torus Knot) — Prior art section. Add after Buniy & Kephart (2005):

Author	Citation	What to Note
Avrin (2012)	Symmetry 4(1), 39–80 — peer-reviewed	Full particle taxonomy from unknot + trefoil. Qualitative topology; no quantitative predictions of fundamental constants.
Duffield (2016–2020)	physicsdetective.com — blog, not peer-reviewed	Independent qualitative proposal: proton as trefoil standing wave, electron as unknot, mass ratio ~1837 noted but not derived.
Vit (2023)	HAL hal-04091300 — preprint	Trefoil wire model for proton/neutron magnetic moments (~99.5%). Classical current-loop method, not standing-wave DOF counting. One quantity, not comprehensive.

✓ Prior art convergence is a strength, not a weakness. Multiple independent researchers reaching the same topology from different starting points is evidence the topology is correct. WST is distinguished by what it does with that topology.